Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	MacLean, David A Ph.D.		
Project Title:	Bed Rest and Neurovascular Function		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dmaclean@kent.edu	**Fax:**	FY 330-672-2250
PI Organization Type:	UNIVERSITY	**Phone:**	330-672-0209
Organization Name:	Kent State University		
PI Address 1:	Room 162D Gym Annex		
PI Address 2:	PO Box 5190		
PI Web Page:			
City:	Kent	**State:**	OH
Zip Code:	44242	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	03/13/2000	**End Date:**	09/28/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-range goal of the principal investigator is to examine the control of blood flow in humans after space flight. The goal of this project is to examine neurovascular function in individuals after two weeks of head-down-tilt bed rest (HDBR). HDBR serves as an Earth-based model for the autonomic adjustments seen during space flight. The main hypothesis of this proposal is that peripheral vasoconstrictor responses are impaired with HDBR and this contributes to orthostatic intolerance (OI). We have noted that subjects with OI after HDBR do not effectively increase sympathetic nerve traffic directed to skeletal muscle. In this proposal we will explore the implications of these observations on norepinephrine (NE) release in skeletal muscle. We will utilize the microdialysis technique to measure skeletal muscle interstitial NE during tilt testing before and after HDBR. This state of the art method will give us previously unobtainable information regarding the release of NE from nerves that reside in the interstitial space and innervate the smooth muscle of blood vessels. In those subjects that are OI, we hypothesize that HDBR will result in an attenuated rise in interstitial NE during tilt. We		

	will block the local neuronal uptake of NE (desipramine) as well as measure indices of intraneuronal NE storage capacity. These experiments will provide important data on NE release rates and how this release is influenced by tilt and HDBR. As a countermeasure we will give pyridostigmine (an oral dose of 60 mg) to ½ of the HDBR subjects. Pilot data from our group suggests that this intervention raises vascular resistance at rest and in response to tilt.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will provide valuable information on the mechanisms of orthostatic intolerance, specifically the role of norepinephrine. In addition, this project will continue to expand the utilization of the microdialysis technique and provide avenues to increase its use as an important measurement of interstitial function.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)